

# PLANNING COURSES

IN-PERSON OR REMOTE LEARNING VIA GOOGLE MEET

LAC 1 - TECHNOPOLE EL GHAZELA - SOUSE - SFAX - NABEUL

# WHO WE ARE



**Established in 2020, the Bridge Training Center is a pathway to success for each student, providing the necessary knowledge and skills to excel their personal and professional life in the IT industry.**



**With an expanding portfolio of training courses, our mission is to help students conquer new knowledge, obtain the required hard skills, and enhance their soft skills.**



**We are committed to continuous improvement and incorporating the latest technologies and innovative methodologies to offer a unique learning experience.**

# AVAILABLE TRAININGS

- Full Stack Development (Spring Boot+Angular)
- MERN STACK Development (React+Node)
- Mobile Development (Flutter+Node/SpringBoot)
- Business Intelligence (BI)
- DevOps
- Artificial Intelligence (IA Chatbot)
- Internet of Things (IoT)
- Marketing
- Cyber Security

# COURSE PLAN

## (Full Stack Spring Boot Angular)

### Frontend (1 month)

- Angular
- HTML / CSS / JavaScript
- Bootstrap

### Backend ( 1 month)

- Spring Boot

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

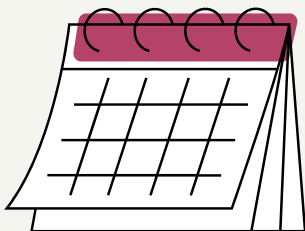
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Duration : 2 months

Number of hours per week : 6h / Week (Online/ Onsite)

Training : 4h / Week (Online/ Onsite)

Supervision : 2h / Week (Online)



### Class Session Time : Supervision Time

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until 13:00**  
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# PLANNING ANGULAR

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 1

- Review of HTML / CSS / JavaScript
- Introduction to Angular Framework
- Setting up work tools
- Start a real world project

## week 2

- Understand the architecture of Angular project
- Explore the concept of routing in Angular
- Master Angular components
- Download a template and explore its layout
- Start Implement in authentication pages

## week 3

- Add filtering options to API endpoints
- Implementing search functionality
- Learn about HTTP and SQL requests

## week 4

- Master Local storage
- Implementing web service integration
- Implement input validation

## Supervision

- Correct use case Diagrams
- Error Handling and Validation
- Monitoring project advancement to ensure completion

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# PLANNING SPRING BOOT

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP

## week 1

- Overview of Spring Boot and its features
- Explain concepts like:  
@RestController,@GetMapping,@PostMapping...
- Setting up development environment
- Develop a basic Spring Boot project

## week 2

- Understanding the structure of a Spring Boot project: packages, folders, and files
- Explanation of basic annotations such as @RestController, @GetMapping, @PostMapping...
- Create API endpoints

## week 3

- Start CRUD operations
- Learn how to delete and retrieve all data
- Connect the project to a database
- Understand the relationship between entities

## week 4

- Understand the concept of JWT
- Enable upload image
- Implement input validation

## Supervision

- Correct class Diagram
- Error Handling and Validation
- Monitoring project advancement to ensure completion

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# COURSE PLAN

## (MERN Stack JS)

### Frontend ( 1 month)

- React Js
- Bootstrap
- Js Query
- ...

### Backend ( 1 month)

- Node JS
- Express JS

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

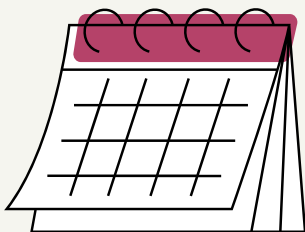
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# PLANNING REACT

1 MONTH : TRAINING +SUPERVISION+INTERNSHIPS



## week 1

- Review of HTML / CSS / JavaScript
- Introduction to React Library
- Setting up work tools
- Start a real world project

## week 2

- Understand the architecture of a React project
- Explore the concept of routing in React
- Master React components
- Download a React template and explore its layout
- Start Implement authentication pages

## week 3

- Create API endpoints and CRUD operations
- Add filtering options to API endpoints
- Implementing search functionality
- Learn about HTTP and SQL requests

## week 4

- Master Local storage and session storage
- Implementing web service integration
- Implement input validation

## Supervision

- Correct use case Diagram
- Error Handling and Validation
- Monitoring project advancement to ensure completion

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# PLANNING NODE

1 MONTH : TRAINING +SUPERVISION+INTERNSHIPS



## week 1

- Introduction to Node JS
- Introduction to Express JS
- Setting up work tools
- Start developing the project



## week 2

- Install Express JS
- Understand the architecture of the project
- Enable upload image/Video/Gif..



## week 3

- Create API endpoints
- Learn how to delete and retrieve all data
- Connect the project to a database
- Understand the relationship between entities



## week 4

- Understand the concept of JWT
- Master local storage
- Implement input validation



## Supervision

- Correct class Diagram
- Error Handling and Validation
- Monitoring project advancement to ensure completion



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# COURSE PLAN

## (Mobile Development)

### Frontend ( 1 month)

- Flutter
- Dart
- ...

### Backend ( 1 month)

- Node JS/ Spring Boot
- Rest API & CRUD
- Back+ Front integration

(Check pages 6 & 9 )

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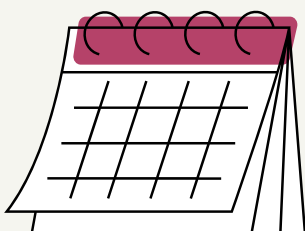
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# PLANNING Flutter

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 1

- Introduction to Flutter
- Getting started with Dart
- Installing Flutter
- Basic architecture of a Flutter application

## week 2

- Advanced widget concepts
- Introduction to navigation
- Interactive widgets
- Introduction to themes and UI customization
- State management in Flutter

## week 3

- Generating ListViews
- Custom components
- Passing data
- Error and exception handling

## week 4

- Local data persistence
- API connection
- Error and exception handling

## Supervision

- Review progress on assigned tasks and provide feedbacks.
- Code Review Sessions: Conduct code reviews to ensure best practices are followed.
- Assist students in troubleshooting issues they encounter during exercises and projects.
- Hold sessions to answer questions, clarify doubts, and ensure understanding of key concepts.

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# COURSE PLAN

## (Business Intelligence)

### Level 1 ( 1 month)

- web scraping
- Python
- talend
- ...

### Level 2 ( 1 month)

- Power BI
- Machine Learning (prediction)
- ...

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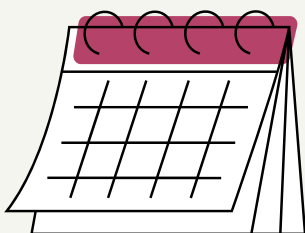
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# PLANNING BI 1

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 1

- Installation of Power BI and Talend.
- Definition of Business Intelligence and its advantages.
- Definition of KPIs and their equivalence tree.
- Web scraping: data collection using two methods (Python and WebScraper extension).
- Descriptive analysis: understanding the data.

## week 2

- Identification of issues.
- Data cleaning with Python using machine learning algorithms.
- Data cleaning with Talend.

## week 3

- Definition of the Kimball method and its advantages.
- Definition of the staging area.
- Extraction, Transformation, and Loading (ETL) of data into the staging area with Talend.

## week 4

- Definition of the Data Warehouse.
- Definition of different decision-making models.
- Extraction, Transformation, and Loading (ETL) of data into the Data Warehouse.

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# PLANNING BI 2

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 5 &6

- Definition of the different components of Power BI.
- Selection of charts for the dashboard.
- Selection of colors for the dashboard.
- Publishing the dashboard on the web.
- Data visualization with Power BI.

## week 7

- Definition of the differences between AI, ML, and DL.
- Use of the CRISP-DM methodology.
- Definition of learning models in machine learning.

## week 8

- Application of machine learning models with Python.
- Selection of the most optimal model.
- Integration of the machine learning component into the dashboard.

## Supervision

- Error Handling, Validation and Installation Phase Supervision.
- Monitoring project advancement to ensure completion
- Supervision Based on Project and Requirements.

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# COURSE PLAN

## (DevOps)

### Level 1 (1 month)

- Docker
- Github
- Jenkins
- Githubactions
- Sonarqube

### Level 2 (1 month)

- Kubernetes
- Argocd
- Ansible
- Terraform
- Prometheus-grafana

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

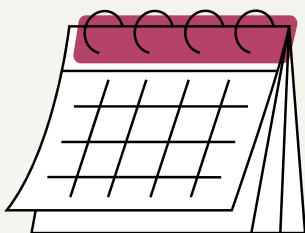
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# PLANNING DevOps 1

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 1

- Overview of DevOps concepts and practices.
- Introduction to Docker: What it is and why it's used.
- Creating a Docker image and running it with Docker commands.
- Introduction to Docker Compose and creating a basic docker-compose.yml file.

## week 2

- Understanding Docker networking and configuring networks.
- Managing Docker volumes for persistent data.
- Introduction to CI/CD with GitHub Actions.
- Creating a simple CI/CD pipeline using GitHub Actions.

## week 3

- Introduction to Jenkins: Setting up and configuring Jenkins.
- Building a Jenkins pipeline.
- Integrating SonarQube for code quality analysis.
- Running the pipeline and analyzing code quality results.

## week 4

- Introduction to Kubernetes and its architecture.
- Setting up a Kubernetes master node.
- Basic commands and concepts in Kubernetes (pods, nodes, clusters).

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# PLANNING DevOps 2

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 5

- Kubernetes Worker Node and Application Deployment
- Adding worker nodes to the Kubernetes cluster.
- Deploying an application in Kubernetes.
- Exposing the application with Kubernetes services.

## week 6

- Understanding and setting up Kubernetes Ingress for routing.
- Configuring Kubernetes volumes for persistent storage.
- Managing node taints and affinities for pod scheduling.

## week 7

- Introduction to GitOps and ArgoCD.
- Setting up ArgoCD for continuous delivery.
- Introduction to Ansible for configuration management and automation.

## week 8

- Advanced Ansible playbooks and roles.
- Introduction to Infrastructure as Code (IaC) with Terraform.
- Using Terraform to provision infrastructure in a cloud environment.

## Supervision

- Review progress on assigned tasks and provide feedbacks.
- Code Review Sessions: Conduct code reviews to ensure best practices are followed.
- Assist students in troubleshooting issues they encounter during exercises and projects.
- Hold sessions to answer questions, clarify doubts...

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# COURSE PLAN

## (Artificial Intelligence)

### Month 1

- Learn how to build Chatbots
- Learn how to use AI tools (NLP,Pandas...)
- ...

### Month 2

- Learn the basics of recommender system
- Design a recommendarion system on python
- ...

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

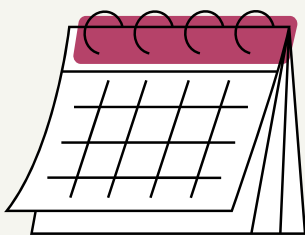
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# COURSE PLAN

(Internet Of Things “IoT”)

## 1 Month

- Introduction to IoT and ESP32
- MQTT protocol
- Interaction with Multiple Sensors and Actuators

...

## Another Months if needed

After completing the IoT, you can continue your learning journey for another month by exploring new frameworks such as Flutter, Node.js, React, or Spring Boot, expanding your skills and diving into software development.

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

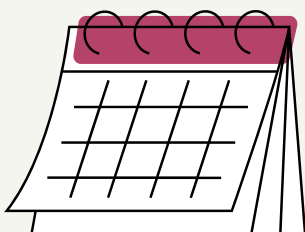
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# PLANNING IOT

1 MONTH : TRAINING +SUPERVISION+INTERNSHIP



## week 1

- Overview of key IoT concepts (connected objects, sensors, actuators).
- Introduction to the ESP32 board:
- Setting up the Arduino IDE to program the ESP32.
- First project: Reading sensors with ESP32
- A simple program to turn an LED on and off connected to the ESP32.

## week 2

- Introduction to MQTT and ESP32 Communication
- Connecting the ESP32 to a Wi-Fi network.
- Programming the ESP32 to send sensor data via MQTT.
- Using an MQTT client (such as MQTT Explorer) to test sending and receiving messages between the ESP32 and the MQTT broker.

## week 3

- Connecting multiple sensors to the ESP32
- Controlling actuators with the ESP32
- Using PWM (Pulse Width Modulation)
- Hardware Interrupts

## week 4

- Setting up a project combining multiple sensors and actuators.
- Implementing alert thresholds
- Integrating an OLED or LCD screen to display sensor data directly from the ESP32.
- Using wireless communication:

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# COURSE PLAN

## (Cyber Security)

### Month 1

- Reconnaissance & Enumeration
- Python Security Automation
- Vulnerability Assessment
- Exploitation Techniques
- Post-Exploitation
- Privilege Escalation
- ....

### Month 2

- Wireless Security Testing
- Social Engineering Attacks
- Defense Evasion Techniques
- Pentest Reporting & CVSS
- Red Team Methodologies
- ....

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

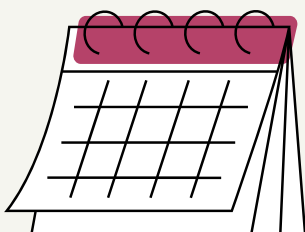
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# PLANNING Cyber Security

2 MONTHS : TRAINING +SUPERVISION+INTERNSHIP

## Month 1

### week 1

- Introduction to Cyber Security
- Understanding Cyber Threats & Attack Vectors
- Setting up a Pentesting Lab Environment
- Kali Linux Essentials
- Networking Fundamentals for Security
- Reconnaissance Technique

### week 2

- Information Gathering & Enumeration
- Automating Reconnaissance using Python
- Port Scanning with Nmap
- Service Fingerprinting
- Vulnerability Assessment
- Web Application Fundamentals

### week 3

- Web Application Penetration Testing
- OWASP Top 10
- Authentication & Session Attacks
- SQL Injection
- Cross-Site Scripting (XSS)
- Introduction to Exploitation

### week 4

- Exploitation & Post-Exploitation
- Privilege Escalation (Windows & Linux)
- Credential Dumping Techniques
- Lateral Movement Concepts
- Custom Exploit Development
- Security Reporting Basics

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# PLANNING Cyber Security

2 MONTHS : TRAINING +SUPERVISION+INTERNSHIP

## Month 2

### week 5

- Wireless Security Fundamentals
- Bluetooth Pentesting
- Wireless Attack Methodologies
- Using Ubertooth & CrackLE
- Wireless Security Assessment
- Mitigation Techniques

### week 6

- Active Directory Security
- Internal Network Penetration Testing
- Pivoting Techniques
- Exploiting Common Misconfigurations
- Advanced Enumeration
- Red Team Methodologies

### week 7

- Defense Evasion Techniques
- Antivirus & EDR Evasion Concepts
- Process Injection Fundamentals
- Command & Control (C2) Basics
- Understanding SOC Detection
- Threat Hunting Perspective

### week 8

- Professional Pentest Reporting
- Executive Summary Writing
- CVSS Scoring
- Technical Documentation
- Presentation of Findings
- Final Project Preparation

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# COURSE PLAN

## (Marketing Digital)

### Month 1

- Market & Competitor Analysis
- Social Media Management
- Graphic Design with Canva
- Video Editing with CapCut Pro
- Content Planning & Scheduling
- Meta Ads Fundamentals
- ...

### Month 2

- Website Creation with WordPress
- Search Engine Optimization (SEO)
- Keyword Research
- On-Page SEO Optimization
- Website Analytics
- SEO Tools & Best Practices
- ...

Face to face course : 🔥 490 DT/Month ➤ ~~600~~ DT/Month

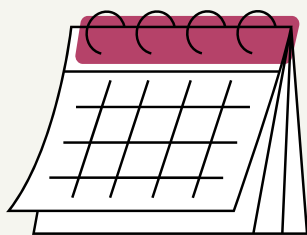
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# PLANNING Marketing Digital

2 MONTH : TRAINING +SUPERVISION+INTERNSHIP

## Month 1

### week 1

- Introduction to Digital Marketing
- Market & Competitor Analysis using AI
- Understanding Customer Personas & Target Audience
- Consumer Behavior & Digital Trends
- Prompt Engineering for Marketers
- Content Research & Idea Generation with AI
- AI Tools for Marketing (ChatGPT, Gemini, Claude)

### week 2

- Content Creation Strategy
- Social Media Management
- Content Calendar Creation
- Personal Branding & Brand Positioning
- Graphic Design with Canva AI
- AI Image Generation for Marketing Campaigns
- Multi-Platform Content Adaptation (Facebook, Instagram, LinkedIn, TikTok)

### week 3

- Storytelling Techniques & Brand Narrative Creation
- Video Editing & Short-Form Content Creation with CapCut AI
- Published Content Performance Analysis

### week 4

- Meta Ads Campaign Creation & Management
- Meta Ads Targeting & Budget Optimization
- Performance Tracking & Campaign Optimization

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# PLANNING Marketing Digital

2 MONTH : TRAINING +SUPERVISION+INTERNSHIP

## Month 2

### week 5

- Introduction to Websites & WordPress
- Domain Name & Web Hosting Basics
- WordPress Installation & Configuration
- Understanding Themes & Plugins
- Creating Pages & Navigation Menus

### week 6

- Introduction to Search Engine Optimization (SEO)
- Keyword Research Fundamentals
- SEO-Friendly Website Structure
- On-Page SEO Optimization
- Creating SEO-Friendly Content
- Introduction to SEO Plugins
- 

### week 7

- Content Marketing
- Website Content Optimization
- Internal Linking
- SEO Tools & Best Practices

### week 8

- Website Analytics & Performance Tracking
- Digital Marketing Campaigns
- SEO Performance Analysis
- Final Practical Project

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# TRAINING PERIODS

## Marketing Digital

Frontend  
(1 month)

Backend  
(1 month)

## Web Development

Frontend  
(1 month)

Backend  
(1 month)

## Artificial Intelligence (IA)

AI Chatbot  
(1 month)

Recommendation  
System (1 month)

## DevOps

Level 1  
(1 month)

Level 2  
(1 month)

## Business Intelligence (BI)

Level 1  
(1 month)

Level 2  
(1 month)

## Internet of Things (IoT)

IoT  
(1 month)

Any other  
Framework  
(1 month/ more)

# TRAINING PERIODS

## Marketing Digital

Creation de  
contenu  
(1 month)

Création site web  
wordpress et seo  
(1 month)

## Cyber Security

Frontend  
(1 month)

Backend  
(1 month)

# TACKLE THE CHALLENGE

**#JOIN US !**



level 1, Lac 1 Tunis / Rockets, Sahloul Sousse



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